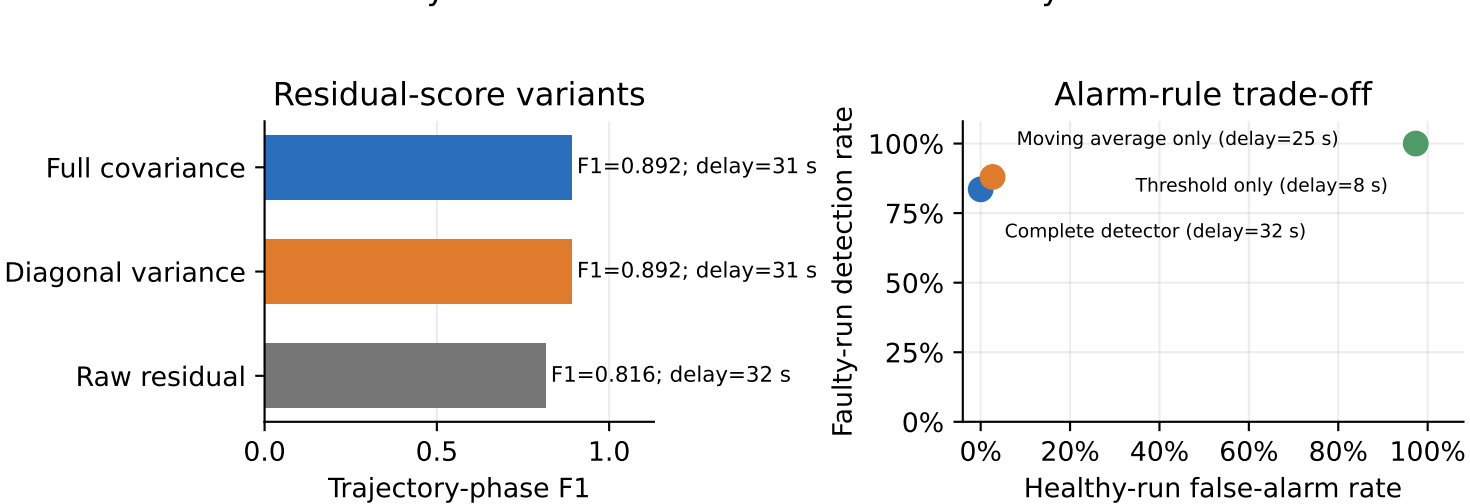
Synthetic benchmark: ablation summary
Residual-score variants
Full covariance
Diagonal variance
Raw residual
F1=0.892; delay=31 s
F1=0.892; delay=31 s
F1=0.816; delay=32 s
0.0
0.5
1.0
Trajectory-phase F1
Alarm-rule trade-off
Faulty-run detection rate
100%
75%
50%
25%
0%
Moving average only (delay=25 s)
Threshold only (delay=8 s)
Complete detector (delay=32 s)
0%
20%
40%
60%
80%
100%
Healthy-run false-alarm rate